Power BI — DAX Measures Library

Measure Organization

Create every item below as a separate **New Measure**. For clean organization, keep measures in a dedicated **_Measures** table.

Important: AOV and Units Sold are separate measures. Do not paste them into the same DAX expression.

SECTION A — Revenue & Sales

Dashboard 1: CEO / Executive

Gross Revenue

```
Gross Revenue =
SUMX(
    order_items,
    order_items[quantity] * order_items[unit_price]
)
```

Net Revenue

```
Net Revenue =
CALCULATE(
    SUMX(
        order_items,
        order_items[line_total] * (1 - RELATED(orders[discount]))
    ),
    orders[order_status] = "Completed"
)
```

Total Orders

```
Total Orders =
CALCULATE(
    DISTINCTCOUNT(orders[order_id]),
    orders[order_status] = "Completed"
)
```

AOV

```
AOV =
DIVIDE(
    [Net Revenue],
    [Total Orders],
    0
)
```

Units Sold

```
Units Sold =
CALCULATE(
    SUM(order_items[quantity]),
    orders[order_status] = "Completed"
)
```

Total Customers

```
Total Customers =
DISTINCTCOUNT(customers[customer_id])
```

Total Refunds

```
Total Refunds =
SUM(returns[refund_amount])
```

Gross Margin

```
Gross Margin =
VAR TotalCost =
    SUMX(
        order_items,
        order_items[quantity] * RELATED(products[cost])
    )
RETURN
    [Net Revenue] - TotalCost
```

Margin %

```
Margin % =
DIVIDE(
    [Gross Margin],
    [Net Revenue],
    0
)
```

Time Intelligence

Requires an active relationship between date_table[Date] and the relevant date field.

Revenue MoM %

```
Revenue MoM % =
VAR CurrentRev = [Net Revenue]
VAR PrevMonthRev =
    CALCULATE(
        [Net Revenue],
        DATEADD(date_table[Date], -1, MONTH)
    )
RETURN
    DIVIDE(CurrentRev - PrevMonthRev, PrevMonthRev, 0)
```

Revenue YoY %

```
Revenue YoY % =
VAR CurrentRev = [Net Revenue]
VAR PrevYearRev =
    CALCULATE(
        [Net Revenue],
        SAMEPERIODLASTYEAR(date_table[Date])
    )
RETURN
    DIVIDE(CurrentRev - PrevYearRev, PrevYearRev, 0)
```

Revenue MTD

```
Revenue MTD =
CALCULATE(
    [Net Revenue],
    DATESMTD(date_table[Date])
)
```

Revenue YTD

```
Revenue YTD =
CALCULATE(
    [Net Revenue],
    DATESYTD(date_table[Date])
)
```

SECTION B — Customer Metrics

Dashboard 1 & 2

New Customers

```
New Customers =
CALCULATE(
    DISTINCTCOUNT(customers[customer_id]),
```

```
    USERELATIONSHIP(customers[signup_date], date_table[Date])
)
```

Repeat Customers

```
Repeat Customers =
VAR CustomerOrderCounts =
    SUMMARIZE(
        FILTER(orders, orders[order_status] = "Completed"),
        orders[customer_id],
        "OrderCount", CALCULATE(DISTINCTCOUNT(orders[order_id]))
    )
RETURN
    COUNTROWS(FILTER(CustomerOrderCounts, [OrderCount] > 1))
```

Repeat Purchase Rate %

```
Repeat Purchase Rate % =
DIVIDE(
    [Repeat Customers],
    [Total Customers],
    0
)
```

Avg Customer Lifetime (Days)

```
Avg Customer Lifetime (Days) =
VAR CustomerLifetimes =
    SUMMARIZE(
        FILTER(orders, orders[order_status] = "Completed"),
        orders[customer_id],
        "Lifetime",
            DATEDIFF(
                CALCULATE(MIN(orders[order_date])),
                CALCULATE(MAX(orders[order_date])),
                DAY
            )
    )
RETURN
    AVERAGEX(CustomerLifetimes, [Lifetime])
```

CLV (Avg)

```
CLV (Avg) =
AVERAGEX(
    SUMMARIZE(customers, customers[customer_id]),
    [Net Revenue]
)
```

SECTION C — CAC & Marketing ROI

Dashboard 3: Growth

Marketing Spend

```
Marketing Spend =
SUM(marketing_campaigns[spend])
```

Total Impressions

```
Total Impressions =
SUM(marketing_campaigns[impressions])
```

Total Clicks

```
Total Clicks =
SUM(marketing_campaigns[clicks])
```

Total Conversions

```
Total Conversions =
SUM(marketing_campaigns[conversions])
```

CTR %

```
CTR % =
DIVIDE([Total Clicks], [Total Impressions], 0)
```

CPC

```
CPC =
DIVIDE([Marketing Spend], [Total Clicks], 0)
```

Conversion Rate %

```
Conversion Rate % =
DIVIDE([Total Conversions], [Total Clicks], 0)
```

CAC

```
CAC =
DIVIDE([Marketing Spend], [New Customers], 0)
```

ROAS

```
ROAS =
DIVIDE([Net Revenue], [Marketing Spend], 0)
```

CLV to CAC Ratio

```
CLV to CAC Ratio =
DIVIDE([CLV (Avg)], [CAC], 0)
```

SECTION D — Funnel & Conversion

Dashboard 3: Growth

Sessions with Page View

```
CALCULATE(DISTINCTCOUNT(web_events[session_id]), web_events[event_name] = "page_view", web_events[
session_id] <> "UNKNOWN_SESSION")
```

Sessions with Product View

```
CALCULATE(DISTINCTCOUNT(web_events[session_id]), web_events[event_name] = "product_view", web_even
ts[session_id] <> "UNKNOWN_SESSION")
```

Sessions with Add to Cart

```
CALCULATE(DISTINCTCOUNT(web_events[session_id]), web_events[event_name] = "add_to_cart", web_event
s[session_id] <> "UNKNOWN_SESSION")
```

Sessions with Checkout

```
CALCULATE(DISTINCTCOUNT(web_events[session_id]), web_events[event_name] = "checkout", web_events[s
ession_id] <> "UNKNOWN_SESSION")
```

Sessions with Purchase

```
CALCULATE(DISTINCTCOUNT(web_events[session_id]), web_events[event_name] = "purchase", web_events[s
ession_id] <> "UNKNOWN_SESSION")
```

Overall Funnel Conversion %

```
DIVIDE([Sessions with Purchase], [Sessions with Page View], 0)
```

SECTION E — RFM & Churn

Dashboard 2: Customer

Use the Python-generated rfm_segments.csv and customer_churn_risk_scores.csv outputs. Relate both to customers using customer_id.

Champions Count

```
CALCULATE(DISTINCTCOUNT(rfm_segments[customer_id]), rfm_segments[rfm_segment] = "Champions")
```

High Risk Customers

```
CALCULATE(DISTINCTCOUNT(customer_churn_risk_scores[customer_id]), customer_churn_risk_scores[risk_
segment] = "High Risk")
```

Avg Churn Probability

```
AVERAGE(customer_churn_risk_scores[churn_probability])
```

SECTION F — Product Metrics

Dashboard 4: Product

Product Revenue

```
Product Revenue =
[Net Revenue]
```

Return Count

```
Return Count =
COUNTROWS(returns)
```

Return Rate %

```
Return Rate % =
DIVIDE([Return Count], [Units Sold], 0)
```

Avg Discount %

```
Avg Discount % =
AVERAGE(orders[discount])
```

Formatting

• Percentage measures: Format as Percentage with 1–2 decimal places.
• Revenue/currency measures: Format as Currency (■ if applicable).
• Large numbers: use compact display such as K/M where appropriate.

Key Fix Applied

The AOV and Units Sold definitions are now explicitly separated into two independent measures. This prevents the syntax error caused by combining `AOV` and `Units Sold` in one DAX expression.